Chemistry-resolved disagreement between PBE and r^{2}SCAN stability verdicts in an AI-generated materials database

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Abstract

Machine-learning crystal discovery now reports hundreds of thousands of thermodynamically stable materials, but these labels rest on generalized-gradient-approximation density functional theory. GNoME reports that 84% of its PBE-stable predictions retain negative phase-separation energies under the r^{2}SCAN meta-GGA, a single aggregate figure. We resolve that figure by chemistry using the 139,541 GNoME materials released with r^{2}SCAN recalculations. Because the dataset is substitution-generated, these materials are not independent observations: they fall into 8,429 structural-prototype clusters with an intra-cluster correlation of 0.14, a design effect of 3.1, and an effective sample size near 44,000. Flip-rate intervals are cluster bootstrap intervals and regression intervals use a cluster-robust sandwich estimator, which inflates standard errors by 2.8 times. Holding element count fixed, the verdict-flip rate ranges from 4.0% [3.2, 4.8] to 30.7% [27.6, 33.8] across disjoint composition cells, and per element from 1.4% (Ac) to 42.8% (Fe), with all 83 element-level contrasts surviving Benjamini–Hochberg correction. The gradient is not an artefact of the MP2020 Hubbard-U corrections applied to PBE but not to r^{2}SCAN: restricted to chemistries where no U is applied, materials containing Co, Cr, Fe, Mn, Mo, Ni, V or W flip at 21.9%, against 7.2% for materials whose only 3d elements are Sc, Ti, Cu or Zn, indistinguishable from materials with no 3d element at all (7.4%). The iron excess over baseline is smallest in oxides (+5 points), the one class where PBE receives a U correction, and largest in borides, pnictides and intermetallics (+34 to +41 points), where it does not: the correction scheme does not generate the disagreement but partially suppresses it where applied. The underlying quantity is a directional shift, not noise: r^{2}SCAN moves 3d oxides and 3d pnictides toward instability by a mean of +11 and +10 meV/atom and moves plain intermetallics toward stability by −12 meV/atom, with comparable scatter in every cell. Open-shell d content is the strongest observed predictor of disagreement, with a cluster-robust odds ratio of 3.45 [3.13, 3.80]; we do not claim an identified mechanism. The gradient survives within every PBE hull-distance bin, is not a band-gap effect, and is not an artefact of which materials were selected for recomputation; on the Materials Project side, where r^{2}SCAN hull completeness can be measured, the open-shell d excess is unchanged across completeness terciles. A matched comparison against Materials Project entries computed at both levels reverses the aggregate comparison: GNoME flips less often overall (13.0% against 14.9%) but more often within oxide and 3d-containing cells. Allocating a fixed r^{2}SCAN recomputation budget by chemistry recovers 37.1% ± 0.9% of disagreements at a 20% budget against 20.1% for uniform sampling, over 20 cluster-wise splits. All results apply to materials with at most four elements: quinary compositions, which GNoME highlights as a principal novelty, are essentially unvalidated at the meta-GGA level (229 of 177,392 recomputed).

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1 Introduction

Computational materials discovery has shifted from enumerating candidates to generating them. GNoME reported 381,000 new entries on the convex hull for a total of 421,000 stable crystals (Merchant et al., 2023); subsequent releases brought the public dataset to 554,054 entries. Generative models such as MatterGen (Zeni et al., 2025) and expanded databases such as Alexandria (Schmidt et al., 2022) have followed. Every one of those stability verdicts is a density functional theory (DFT) calculation at the Perdew–Burke–Ernzerhof (PBE) level (Perdew et al., 1996), executed in VASP (Kresse and Furthmüller, 1996) with Materials Project-compatible settings (Jain et al., 2013) including the MP2020 empirical anion and Hubbard-U corrections (Wang et al., 2021).

That GGA formation energies carry errors of order 0.1 eV/atom is well established. Kirklin et al. report a mean absolute error of 0.136 eV/atom against 1,670 experimental formation energies, falling to 0.081 eV/atom with elemental fitting, against an experiment-versus-experiment MAE of 0.082 eV/atom (Kirklin et al., 2015). What has not been measured is how those errors distribute across the chemical space that generative discovery actually occupies.

The distinction is practical. A laboratory choosing which predicted compound to attempt cannot act on an aggregate error rate. A database maintainer deciding where to spend a finite higher-fidelity recomputation budget faces the same problem at scale. And a benchmark that treats a PBE hull as ground truth (Riebesell et al., 2025) implicitly assumes the label noise is uniform across chemistry.

GNoME's authors anticipated the question. They recomputed a subset with the r^{2}SCAN meta-GGA (Furness et al., 2020) and reported that 84% of discovered binaries and ternaries retain negative phase-separation energies, with 86.8% of tested quaternaries remaining stable on the r^{2}SCAN hull (Merchant et al., 2023). Supplementary Note 2 observes that the discrepancy concentrates near the hull and that the same effect appears in the Materials Project and OQMD, but gives no breakdown by chemistry. The authors released the underlying r^{2}SCAN data with the statement that it is provided "expectantly serving as a foundation for analysing discrepancies between functional choices."

1.1 What this paper measures, and what it does not

We measure *disagreement between two computational methods*. Three concepts must stay separate:

1. **Functional disagreement.** PBE and r^{2}SCAN assign different stability classifications. This is what we observe.

2. **Computational accuracy.** One functional is closer to the truth for a given material. We do not establish this.

3. **Experimental stability.** The material is thermodynamically stable in reality. We say nothing about this.

We therefore use the term *flip* throughout, and avoid describing flipped labels as "incorrect". Where we discuss practical consequences, the operative quantity is the share of labels that a higher-fidelity recomputation would revise — which is a statement about what the expensive calculation would say, not about physical reality. This distinction constrains every claim that follows, including the budget model in Section 4.12, whose payoff is measured in disagreements surfaced rather than errors corrected.

1.2 Questions and contributions

1. **Q1.** Does the flip rate vary systematically with chemistry, controlling for element count, and what drives the variation?

2. **Q2.** Do AI-discovered materials flip at a different rate than previously catalogued materials in matched chemistry?

3. **Q3.** Given a fixed recomputation budget, how much more disagreement is surfaced by allocating by chemistry rather than uniformly?

Our contributions: (i) a chemistry- and element-resolved map of functional-level stability-verdict disagreement on an AI-generated materials database; (ii) controlled tests showing the gradient survives stratification by hull distance and is not produced by band gap, structure, cell size or recomputation selection, and that the empirical correction scheme partially suppresses rather than creates it; (ii′) the observation that the gradient arises from a chemistry-dependent sign in the PBE-to-r^{2}SCAN energy shift; (iii) a matched comparison against previously catalogued materials showing that the published aggregate comparison conceals a sign reversal; (iv) a verification-budget model evaluated over repeated cluster-wise splits; and (v) the observation that the compositional regime GNoME advertises most prominently is almost entirely unvalidated at the meta-GGA level.

The analysis was pre-registered: decision rules, including the threshold that would have caused us to abandon the chemistry-resolved framing, were committed to version control before any stratified result was computed.

2 Related work

2.1 Functional comparisons on known materials

Kingsbury et al. benchmarked r^{2}SCAN against SCAN and PBEsol across roughly 6,000 solids and found r^{2}SCAN more accurate than either for formation energies (Kingsbury et al., 2022a), and developed the mixing scheme the Materials Project uses to place PBE($+U$) and r^{2}SCAN energies on a single phase diagram (Kingsbury et al., 2022b).

Two results bear directly on what follows. Kothakonda et al. tested r^{2}SCAN thermochemistry on more than 1,000 solids and concluded that for transition metals, intermetallics and weakly bound solids, GGAs are comparable to meta-GGAs, and that for Type 2 decomposition enthalpies PBE is slightly more accurate (Kothakonda et al., 2023). Bartel et al. reached a compatible verdict from the reaction side: across 646 decomposition reactions that are not simply formation reactions, PBE (70 meV/atom) and SCAN (59 meV/atom) perform similarly (Bartel et al., 2019). Both predict, on physical grounds, that intermetallics should be the chemistry in which the two functionals most nearly *agree*. Section 4.1 confirms this on an AI-generated set, which serves as a consistency check on our pipeline as well as a result.

2.2 Correction schemes and their pathologies

The MP2020 scheme quantifies uncertainty in DFT energy corrections and shows that many polymorphs classified as unstable may be stable once that uncertainty is propagated (Wang et al., 2021). Warford, Thiemann and Csányi argue that applying U selectively — only when oxygen or fluorine is present — creates two mutually incompatible potential-energy surfaces within one dataset (Warford et al., 2026). Section 4.3 exploits that boundary as a natural experiment.

2.3 Chemistry-resolved error analysis

Sun et al. data-mined 29,902 observed phases to produce a chemistry-resolved scale of metastability (Sun et al., 2016). Yu et al. map universal MLIP formation-energy error per element across 19,998 unary and binary phases (Yu et al., 2024). Both stratify a computed-energy error by chemistry; neither addresses functional-level verdict flips, and neither concerns a generative dataset.

2.4 Critiques of AI-generated catalogues

Cheetham and Seshadri questioned how many GNoME entries are simultaneously novel, credible and useful (Cheetham and Seshadri, 2024). Liu and Meng report that metals constitute over 60% of GNoME and that Ho, Tb, Rh and Er are over-represented relative to the Materials Project (Liu and Meng, 2024); that compositional skew matters for interpreting aggregate comparisons (Section 4.11). Leeman et al. examined the companion autonomous-synthesis work and concluded that no new materials were demonstrated in it (Leeman et al., 2024); the underlying paper has since been corrected (Szymanski et al., 2023, 2026). None of these addresses label reliability.

2.5 Flip-rate framing

The closest prior framing is Reents, Bercx and Pizzi, whose supplementary material plots a PBEsol-versus-SCAN stability flip rate against threshold for Alexandria (Reents et al., 2026). That analysis is aggregate, concerns a different database and functional pair, serves to validate a machine-learned correction rather than to characterise a generative dataset, reports no chemistry resolution, and gives its numerical values only in a figure. An et al. re-screened an ML-generated set, finding 298 of 603 curated structures remained PBE-stable on complete hulls and 109 of 166 dynamically stable candidates survived a SCAN refinement (An et al., 2026); they excluded 142 rare-earth compounds outright because of f-electron and pseudopotential uncertainty. Huang et al. map PBE formation energies onto the r^{2}SCAN scale across 380,190 paired MatPES entries (Huang et al., 2026; Kaplan et al., 2025) but verify stability verdicts for only six binary systems.

We are not aware of a prior chemistry- or element-resolved verdict-flip analysis on GNoME or any other AI-generated database, but we state this as the result of our own literature search rather than as an established priority claim, and the contribution does not depend on it. Stated without reference to priority, what this paper adds is a chemistry- and element-resolved analysis of functional-level stability-verdict disagreement in the recomputed GNoME subset, with explicit treatment of correlated substitution-generated structures and of recomputation selection, a matched comparison against previously catalogued materials, and an evaluation of chemistry-aware budget allocation. Each component has relatives in the literature cited above; the combination, and its application to a generative dataset, is what is new here.

3 Data and methods

3.1 Datasets

GNoME. Two files from the public release (Merchant et al., 2023): `stable_materials_summary.csv` (554,054 unique material IDs with PBE-level decomposition energies, band gaps, space groups and cell information) and `stable_materials_r2scan.csv` (139,541 unique IDs with r^{2}SCAN recalculations). The release notes state that not all materials remain stable under the r^{2}SCAN metric, and we confirm this: positive decomposition energies are present throughout, so the file is not censored

to survivors. This matters because the corresponding Materials Project release of GNoME entries appears to have been selected on r^{2}SCAN stability and cannot serve the same purpose.

Each file reports decomposition energy against several reference sets: against the Materials Project hull, against MP and OQMD jointly, and against a set the release labels "All" and does not further define. We take the MP-referenced column as primary because it reproduces the published aggregate, report the MP+OQMD variant as a sensitivity check, and examine the "All" column separately in Section 4.14 because it gives a materially different result.

Of the 117,043 recomputed materials that can be matched to a PBE decomposition energy in the summary file, 25 have a positive PBE value and were therefore not stable to begin with. They cannot flip and are excluded from every analysis that uses PBE energies. They are 0.02% of the set and no reported number changes.

Known materials. We query the Materials Project thermo endpoint for entries carrying both `GGA_GGA+U` and `r2SCAN` thermodynamic data. This enumeration value is case-sensitive: the string `R2SCAN` returns zero documents without raising an error, a silent failure mode that would make a replication attempt look like a refutation. We retrieved 37,123 r^{2}SCAN and 152,426 GGA+U documents; 35,176 material IDs carry both, of which 22,756 are stable at the GGA+U level and therefore capable of flipping. The `r2SCAN` thermo type is distinct from the Materials Project's mixed GGA/GGA+U/r^{2}SCAN scheme (Materials Project, 2026): its hull is built from whichever r^{2}SCAN entries exist in a chemical system, with no requirement that every competing phase has been recomputed. We measure how complete those hulls are in Section 4.14.

3.2 Definitions

For GNoME a material is *flipped* when its r^{2}SCAN decomposition energy exceeds zero. For Materials Project entries, where `energy_above_hull` is non-negative by construction, we define stability as $E_{\text{hull}} \leq 10^{-6}$ eV/atom and flipping as GGA+U-stable with $E_{\text{hull}}^{\text{r}^2\text{SCAN}} > 10^{-6}$.

We classify each composition into one of seven mutually exclusive anion classes by priority: oxide (contains O); halide (F, Cl, Br or I, no O); chalcogenide (S, Se or Te, none of the above); pnictide (N, P, As, Sb or Bi, none of the above); hydride (H, none of the above); boride/carbide/silicide (B, C or Si, none of the above); intermetallic (none of the above). We flag the presence of a lanthanide (La–Lu) and of a 3d transition metal (Sc–Zn). Cells are the Cartesian product of element count, anion class, lanthanide flag and 3d flag, and are disjoint by construction. Only cells with $n \geq 500$ enter the primary analysis; per-element analyses use $n \geq 300$.

3.3 Materials are not independent observations

GNoME is substitution-generated: the authors attribute 232,477 of 381,000 stable structures to symmetry-aware partial substitution (Merchant et al., 2023). Materials sharing a parent structure and differing by one or two ionic substitutions are not independent samples of chemical space, and may share relaxation behaviour, competing phases and systematic error.

We quantify this using a structural-prototype proxy: the pair (space group number, anonymised stoichiometry), where anonymised stoichiometry replaces element identities with sorted site counts, so that $Nd_7Os_1Pr_3Si_5$ becomes A1B3C5D7. The 139,541 recomputed materials fall into **8,429 such clusters**, mean size 16.6. The five largest hold 2,953, 2,711, 2,374, 1,570 and 1,122 materials. Only 29.6% of clusters are singletons, and 87.5% of all materials sit in clusters of ten or more.

The intra-cluster correlation of the flip outcome is $\rho = 0.138$ (ANOVA estimator). With mean cluster size $n_0 = 16.5$ the design effect is $1 + (n_0 - 1)\rho = 3.14$, implying an effective sample size

near 44,000 rather than 139,541, and confidence intervals wider by a factor of about $\sqrt{3.14} = 1.77$ than naive binomial intervals.

All intervals in this paper are therefore cluster bootstrap intervals: we resample prototype clusters with replacement, not individual materials, 600 replicates for cells and 300 for elements. Across the 40 primary cells the observed widening relative to naive intervals has median 1.82 and ranges from 1.14 to 4.65. An earlier version of this analysis used naive intervals and overstated precision accordingly; the point estimates are unchanged.

The prototype proxy is a choice, so we test an alternative. Clustering on anonymised stoichiometry alone, ignoring space group, is far coarser: 2,186 clusters, mean size 63.8, largest 7,007. The headline quaternary contrast is essentially unaffected: 30.7% [27.3, 34.2] against 4.0% [3.2, 4.7] under the coarse definition, compared with 30.7% [27.7, 33.6] against 4.0% [3.3, 4.8] under the primary one (both from a 500-replicate re-draw; the 600-replicate primary intervals in Appendix A differ in the last digit). A third definition adding the element set is too fine to be informative (mean cluster size 1.03) and we do not use it. We report the primary definition throughout and note that results are not sensitive to coarsening it.

3.4 Multiple comparisons

The per-element analysis performs 83 tests. We report Benjamini–Hochberg adjusted p-values for the contrast between each element's flip rate and the global rate, computed from the cluster bootstrap standard error. The primary cell-level comparison and the lanthanide hypothesis were pre-registered; the per-element analysis was not, and is labelled exploratory throughout despite surviving correction.

3.5 Uncertainty for the regression

Logistic regression coefficients are fitted by iteratively reweighted least squares. Standard errors use a cluster-robust sandwich estimator, $V = B\left(\sum_g X_g^\top u_g u_g^\top X_g\right) B$ with $B = (X^\top W X)^{-1}$ and u_g the score contributions of cluster g, clustering on the same structural prototypes used for the bootstrap. Relative to naive analytic errors this inflates the standard error on the open-shell d coefficient by a factor of 2.77, consistent with the design effect estimated independently in Section 3.3.

3.6 Pre-registration

The pre-registered primary rule was that a spread of ≥ 5 percentage points between the highest and lowest cell, with non-overlapping intervals at fixed element count, would support a standalone analysis; below 3 points the result would be reported as a single parameter elsewhere. The secondary pre-registered claim was that lanthanide-containing materials would flip at or below the global rate. That claim was partially refuted and we report it as such. Rules and results were committed in that order.

4 Results

4.1 The aggregate figure reproduces, and conceals an order of magnitude of variation

Across all 139,541 recomputed materials the flip rate is 13.04% against the Materials Project hull, 15.93% against MP+OQMD, and 25.8% against the most inclusive reference set. The first repro-

duces the published 86.8% quaternary survival figure to within 0.2 percentage points, validating the pipeline against the source before any new claim is made.

Resolved by chemistry at fixed element count, the picture changes entirely (Figure 1; full table with cluster-robust intervals in Appendix A). Among quaternary materials, 23 cells qualify and the flip rate ranges from 4.0% [3.2, 4.8] for plain intermetallics containing neither a lanthanide nor a $3d$ metal ($n = 5{,}025$) to 30.7% [27.6, 33.8] for $3d$-containing oxides without a lanthanide ($n = 2{,}478$). The spread of 26.7 points is more than five times the pre-registered threshold and the intervals remain disjoint under clustering. Ternaries show a comparable spread of 23.9 points across 17 cells.

The ordering matches what the functional-benchmarking literature predicts. Kothakonda et al. find GGAs comparable to meta-GGAs precisely for transition metals and intermetallics (Kothakonda et al., 2023), and intermetallics are where we measure the lowest flip rate; Bartel et al. reach a similar conclusion from decomposition reactions (Bartel et al., 2019). Our contribution is to show that this known accuracy structure propagates into a *verdict* disagreement of practical magnitude on a generative dataset, and to quantify it cell by cell.

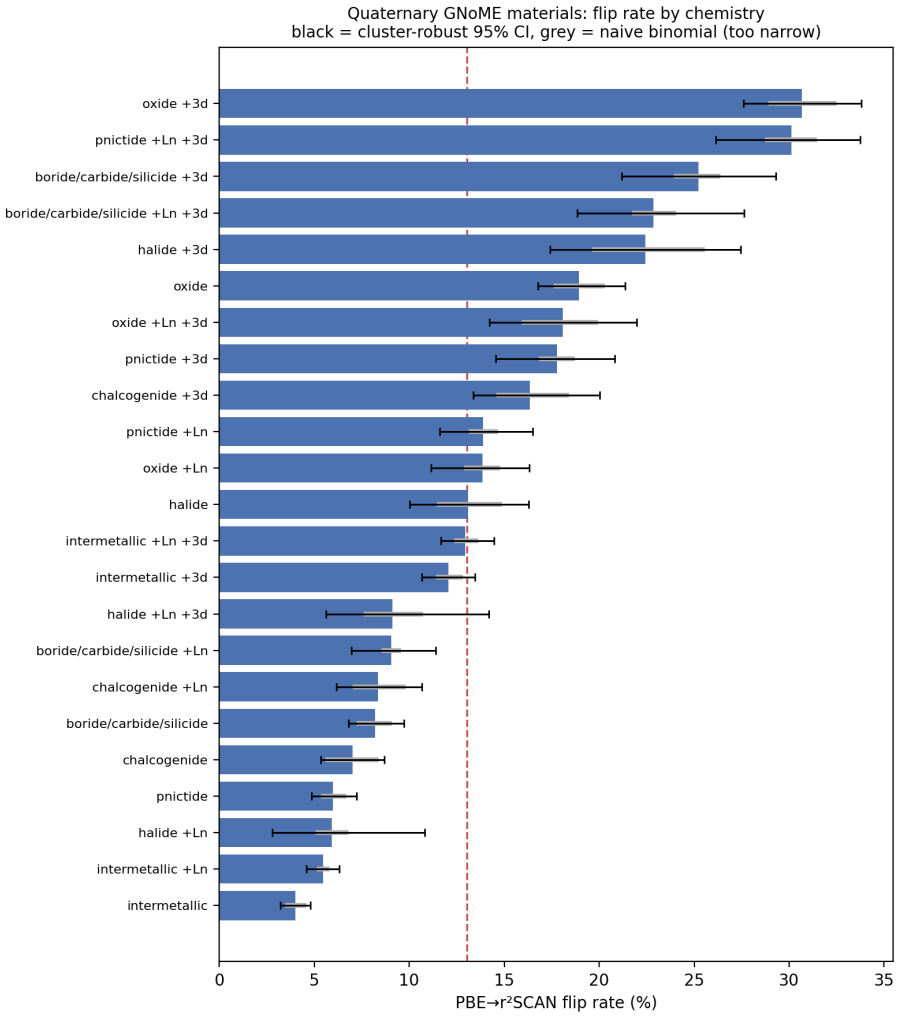


Figure 1: Flip rate by composition cell for quaternary GNoME materials. Cells are disjoint. Black bars are cluster bootstrap 95% intervals; the grey bars behind them are the naive binomial intervals, shown to make the size of the correction visible. The dashed line marks the global rate.

4.2 Per-element rates span a factor of thirty, sevenfold among non-f elements

Computing the flip rate for every element appearing in at least 300 recomputed materials (83 elements; Figure 2, full table in Appendix B) gives a sharper picture. Iron-containing materials flip at 42.8% [40.2, 46.2] (n = 5,762, 1,218 prototype clusters). Cobalt follows at 33.2% [30.0, 35.6], chromium at 30.7% [25.6, 36.8] and manganese at 25.2% [22.8, 27.9]. At the other end, actinium-containing materials flip at 1.4% [1.0, 2.0], protactinium at 3.6% [2.6, 4.8], magnesium at 5.7% [4.2, 7.7] and iridium at 6.2% [5.5, 7.1].

All 83 element-level contrasts against the global rate remain significant after Benjamini–Hochberg correction at the 0.05 level. This is unsurprising given cluster counts in the hundreds to thousands per element, and we note it to close the multiple-comparisons question rather than as evidence of anything in itself. We continue to treat the per-element analysis as exploratory because it was not pre-registered.

Three cautions apply to this ranking. Per-element rates are not disjoint: an iron oxide counts toward both Fe and O, so an element's rate is confounded with the anion classes in which it tends to occur. Section 4.3 addresses this by reporting the *excess* flip rate attributable to Fe, Co and Ni within each anion class, which is the quantity to read for element-level conclusions. The bottom of the ranking is dominated by actinides (Ac, Pa, Np), which carry the same frozen-f pseudopotential caveat as lanthanides (Section 4.10), so the "thirtyfold" range should not be over-read; the range among non-f elements is roughly sevenfold. And the correspondence with the Materials Project U list is imperfect: nickel sits at 13.4%, essentially the global average. Section 4.3 shows that nickel's excess is concentrated in anion classes where it is more likely to carry a magnetic moment, which is consistent with magnetic open-shell character rather than U-list membership being the operative variable.

4.3 The correction scheme suppresses the disagreement where applied; it does not create it

PBE labels carry the MP2020 anion and Hubbard-U corrections; raw r^{2}SCAN carries none. If the gradient simply tracked where U is applied, it would be an artefact of bookkeeping. The Materials Project applies U only to Co, Cr, Fe, Mn, Mo, Ni, V and W, and only in oxides and fluorides (Warford et al., 2026), which partitions the dataset into a corrected and an uncorrected regime and lets us test the question two ways.

First test: restrict to the uncorrected regime. Among compositions containing neither oxygen nor fluorine, where no U is applied under either functional, quaternary materials containing one of the eight U-list elements flip at 21.9%, those whose only 3d elements are Sc, Ti, Cu or Zn at 7.2%, and those with no 3d element at 7.4% (Table 1). The ternary subset shows the same three-way separation. The effect is present where no correction exists, so the correction does not generate it.

Second test: compare the corrected and uncorrected regimes directly. The sharper question is how much iron raises the flip rate *within* each anion class (Table 2). The excess is smallest in oxides, where PBE+U is applied, and largest in borides, pnictides and intermetallics, where PBE runs with no correction at all. Cobalt shows the same pattern. Nickel, which is frequently low-spin or non-magnetic in metallic environments, shows essentially no excess in intermetallics and a modest one in oxides and chalcogenides.

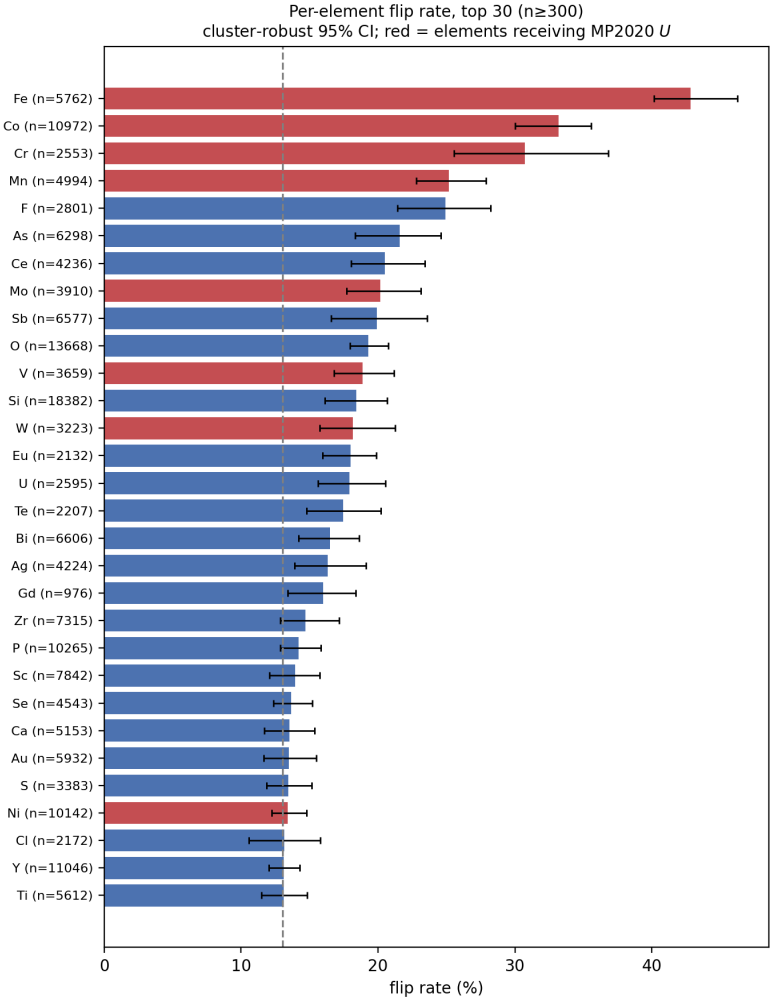


Figure 2: Per-element flip rate, 30 highest of 83 elements with $n \geq 300$, with cluster-robust 95% intervals. Red bars mark the eight elements receiving an MP2020 Hubbard U in oxides and fluorides.

Table 1: Flip rate outside oxides and fluorides, where no Hubbard U is applied under either functional.

Element count	Group	n	Flip rate
Quaternary	Contains Co, Cr, Fe, Mn, Mo, Ni, V or W	29,482	21.9%
Quaternary	Only Sc, Ti, Cu or Zn among 3d metals	14,649	7.2%
Quaternary	No 3d element	54,795	7.4%
Ternary	Contains Co, Cr, Fe, Mn, Mo, Ni, V or W	5,245	28.4%
Ternary	Only Sc, Ti, Cu or Zn among 3d metals	3,102	10.8%
Ternary	No 3d element	15,765	9.6%

Table 2: Excess flip rate attributable to the presence of Fe, Co or Ni, within each anion class. "Without" is the flip rate of materials in the same class lacking that element. Classes with fewer than 100 element-containing materials omitted.

Anion class	n with Fe	With Fe	Without Fe	Excess (pp)
boride/carbide/silicide	1,268	54.0%	13.0%	+41.0
pnictide	1,264	50.3%	15.0%	+35.3
intermetallic	1,904	42.2%	8.0%	+34.2
chalcogenide	154	41.6%	11.5%	+30.0
halide	476	24.4%	10.9%	+13.5
oxide	610	24.1%	19.1%	+5.0
Co excess: boride +34.0, intermetallic +22.2, pnictide +22.0, oxide +4.6, halide −0.7				
Ni excess: chalcogenide +14.2, oxide +10.3, intermetallic +2.0, pnictide −1.5, boride −2.2				

Reading. Together the two tests support a more specific conclusion than "not the correction scheme." The correction scheme does not generate the disagreement: it is present, and largest, where no correction is applied. But where U *is* applied, in oxides, the open-shell d excess shrinks from over thirty points to five. The empirical correction is doing part of what the meta-GGA does. The largest PBE/r^{2}SCAN disagreements in the entire dataset are open-shell $3d$ elements in non-oxide environments, which is precisely the regime where PBE is run with no correction and where Warford et al. identify the uncorrected potential-energy surface (Warford et al., 2026).

Scandium and titanium are d^0 in their common oxidation states and copper and zinc are d^{10}; all four behave like main-group elements. Nickel's weak effect, concentrated in anion classes where it is more likely to carry a moment, suggests the operative variable is magnetic open-shell d character rather than membership of the U list as such. We state this as the most consistent reading of the association, not as an identified mechanism.

4.4 The gradient survives within every PBE hull-distance bin

The most obvious alternative explanation for Section 4.1 is proximity. Materials close to the PBE hull need only a small energy change to cross it. If $3d$ oxides sit nearer the hull than intermetallics, they will flip more often for no reason connected to the functional. Supplementary Note 2 of the source work reports that disagreement concentrates near the hull, which makes this a live concern. Testing it was pre-registered as robustness gate 6; we report it here.

Table 3 stratifies quaternary materials by PBE decomposition energy. Two things are visible. Flip rate rises steeply toward the hull in every cell, confirming the Note 2 observation: plain intermetallics go from 0.5% in the $[-0.1, -0.05]$ band to 15.9% in $[-0.01, 0]$. And the chemistry gradient is present inside every band. In the $[-0.05, -0.02]$ band, $3d$ oxides flip at 23.1% and plain intermetallics at 1.2%; in $[-0.01, 0]$, 40.1% against 15.9%. The within-band spread between those two cells is 10.6, 21.8, 25.2 and 24.2 points across the four populated bands.

The cells are not differently positioned relative to the hull. Median PBE decomposition energy is −20.8 meV/atom for $3d$ oxides, −22.4 for plain intermetallics, −27.3 for oxides without $3d$ and −28.1 for $3d$ pnictides. Hull proximity and chemistry act as roughly independent factors, and the chemistry effect is not a proximity effect in disguise.

Table 3: Quaternary flip rate by PBE decomposition energy band and composition cell. Cells with fewer than 200 materials in a band are blank. Counts in parentheses.

PBE band (eV/atom)	oxide+3d	oxide	pnictide+3d	pnictide	interm.+3d	interm.
< -0.1	—	3.9% (563)	5.5% (494)	0.5% (794)	2.0% (402)	0.8% (930)
$[-0.1, -0.05]$	11.1% (443)	6.1% (1063)	14.6% (1748)	1.3% (2364)	3.2% (1987)	0.5% (2710)
$[-0.05, -0.02]$	23.1% (1070)	15.5% (2132)	19.8% (3608)	8.2% (3810)	7.9% (6709)	1.2% (7777)
$[-0.02, -0.01]$	29.4% (744)	24.3% (1159)	28.3% (1752)	20.8% (1970)	16.0% (3947)	4.3% (4512)
$[-0.01, 0]$	40.1% (802)	27.8% (1142)	37.7% (1493)	27.0% (1442)	27.2% (3263)	15.9% (4937)

4.5 r^{2}SCAN shifts chemistries in opposite directions

A flip is a binary summary of a continuous quantity: $\Delta = E_{\text{dec}}(\text{r}^2\text{SCAN}) - E_{\text{dec}}(\text{PBE})$. Examining Δ directly shows that the flip-rate gradient is a consequence of a directional, chemistry-dependent shift rather than of chemistry-dependent noise (Table 4).

Table 4: Change in decomposition energy from PBE to r^{2}SCAN, quaternary materials. Positive values mean r^{2}SCAN moves the material toward instability.

Cell	n	mean Δ (meV/at)	median	s.d.	share $\Delta > 0$
oxide + 3d	3,178	+11.3	+4.8	52.6	60%
pnictide + 3d	9,095	+9.6	+3.6	41.8	60%
oxide	6,059	+3.6	+1.5	45.1	50%
pnictide	10,380	−1.8	−0.8	39.0	50%
intermetallic + 3d	16,308	−1.8	−1.0	33.1	50%
intermetallic	20,866	−12.4	−4.8	36.0	30%

The scatter of Δ is comparable across cells, 33 to 53 meV/atom. What differs is the mean. r^{2}SCAN systematically destabilises 3d oxides and 3d pnictides relative to PBE by about 10 meV/atom, and systematically *stabilises* plain intermetallics by about 12 meV/atom; only 30% of plain intermetallics move upward at all. Flips occur where an upward shift carries a near-hull material across zero, which is why they concentrate in the cells with positive mean Δ.

This sharpens the connection to prior benchmarking. Kothakonda et al. and Bartel et al. find GGA and meta-GGA "comparable" for intermetallics (Kothakonda et al., 2023; Bartel et al., 2019). Our data agree that the scatter is comparable, and add that the mean differs in sign between chemistries. Whether r^{2}SCAN's stabilisation of intermetallics or its destabilisation of 3d oxides is the more accurate direction is a question about experiment that this paper does not address.

4.6 Nor is it a band-gap effect

The GNoME release includes a PBE band gap for 114,723 recomputed materials. Marginally, the relationship is non-monotonic (Figure 3): metals ($E_g \leq 0.01$ eV) flip at 13.6%, small-gap materials (0.01–0.5 eV) at 9.6%, and the rate then climbs to 21.2% above 3 eV. A "metals are harder" story is wrong on its face.

More decisively, the apparent dependence largely disappears once 3d content is controlled for (Table 5). Among ternary 3d-containing materials, metals flip at 24.0% and non-metals at 22.6%; among quaternaries, 19.4% against 17.6%. Among materials with no 3d element the two differ by about one point in the opposite direction.

Table 5: Flip rate by metallicity, stratified by $3d$ content. Metal defined as PBE band gap ≤ 0.01 eV.

Element count	$3d$ present	Metal	n	Flip rate
Ternary	yes	yes	4,714	24.0%
Ternary	yes	no	1,442	22.6%
Ternary	no	yes	8,367	11.0%
Ternary	no	no	4,908	11.9%
Quaternary	yes	yes	29,113	19.4%
Quaternary	yes	no	10,423	17.6%
Quaternary	no	yes	36,707	8.3%
Quaternary	no	no	18,655	9.5%

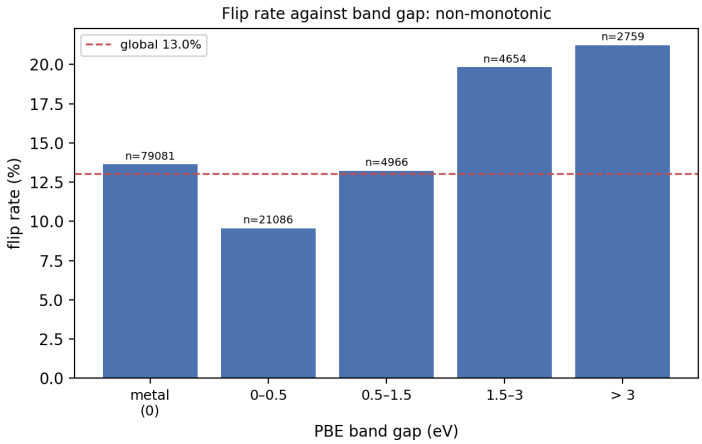


Figure 3: Flip rate against PBE band gap. The relationship is non-monotonic, and Table 5 shows it is largely mediated by $3d$ content.

4.7 Nor is it an artefact of which materials were recomputed

The recomputed subset is not a random sample. We model the probability of recomputation directly, fitting a logistic model of recomputation on composition, element count, cell size and PBE decomposition energy (linear and quadratic) across the 371,870 ≤4-element materials, of which 116,814 (31.4%) were recomputed.

Selection is indeed chemistry-dependent: oxides are more likely to be recomputed (odds ratio 1.64), as are borides, carbides and silicides (1.58) and pnictides (1.43), while lanthanide-containing materials (0.68) and larger cells (0.65 per standard deviation of log sites) are less likely. The referee's concern is therefore well founded in its premise.

It does not, however, change the result. Inverse-probability weighting moves the overall ≤4-element flip rate from 13.37% to 12.71%, and the spread across quaternary cells from 22.0 to 21.9 percentage points (Table 6). Cell-level estimates move by at most 2.4 points and the ordering is preserved.

Table 6: Quaternary cells, naive against inverse-propensity-weighted flip rate. Selected rows; full table in the repository.

Cell	n	Naive	IPW
oxide, $3d$	3,178	27.2%	27.2%
boride/carbide/silicide, $3d$	7,947	25.2%	25.8%
pnictide, $3d$	9,097	22.6%	24.6%
oxide, no $3d$	6,059	16.8%	16.7%
intermetallic, $3d$	16,318	13.0%	13.1%
halide, no $3d$	4,302	7.2%	8.1%
intermetallic, no $3d$	20,871	5.3%	5.3%
Spread		**22.0 pp**	**21.9 pp**

Inverse-probability weighting corrects only for selection on observed features. If the authors selected materials for recomputation on something we cannot see — convergence behaviour, internal confidence scores, manual interest — that remains uncorrected. What we can say is that the chemistry gradient is not generated by any of the selection mechanisms we can model.

4.8 Structural and size effects are small

By crystal system, flip rates run from 16.8% (triclinic) to 11.1% (hexagonal), with cubic at 14.3% and the rest between 12.6% and 13.3%. The triclinic excess is plausibly a proxy for low-symmetry structures being harder to relax consistently, but we have not established that. By cell size, rates rise from 11.3% (≤10 sites) to 16.3% (>32 sites); in the regression below the size term is not significant once composition is controlled for.

4.9 A logistic model over composition

Table 7 reports a logistic regression of flip probability on composition features across all 139,541 materials, solved by iteratively reweighted least squares.

Three readings, and three cautions.

The presence of an open-shell d element multiplies the *odds* of a flip by 3.45. An odds ratio is not a risk ratio: at the reference baseline of about 6% the corresponding probability ratio is near

Table 7: Logistic regression of flip probability. Reference category is an intermetallic with no $3d$ metal and no lanthanide. $n = 139{,}541$; McFadden pseudo-$R^2 = 0.066$. Intervals are cluster-robust (sandwich estimator on structural prototypes). The final column gives the inflation of the standard error relative to the naive analytic value.

Feature	Coefficient	Odds ratio	Cluster-robust 95% CI	SE inflation
Intercept	−2.780	0.062	[0.056, 0.069]	2.24×
Open-shell d (U list)	+1.238	3.448	[3.129, 3.799]	2.77×
Other $3d$ (Sc, Ti, Cu, Zn)	+0.096	1.101	[0.984, 1.232]	1.98×
Lanthanide present	+0.023	1.023	[0.933, 1.121]	2.74×
Oxide	+0.903	2.467	[2.161, 2.815]	2.42×
Pnictide	+0.666	1.946	[1.700, 2.228]	3.00×
Boride/carbide/silicide	+0.518	1.678	[1.437, 1.959]	3.32×
Chalcogenide	+0.336	1.399	[1.205, 1.624]	1.97×
Halide	+0.286	1.330	[1.047, 1.691]	3.42×
Element count (z-scored)	−0.177	0.838	[0.807, 0.870]	2.32×
log sites (z-scored)	+0.015	1.015	[0.958, 1.075]	3.45×

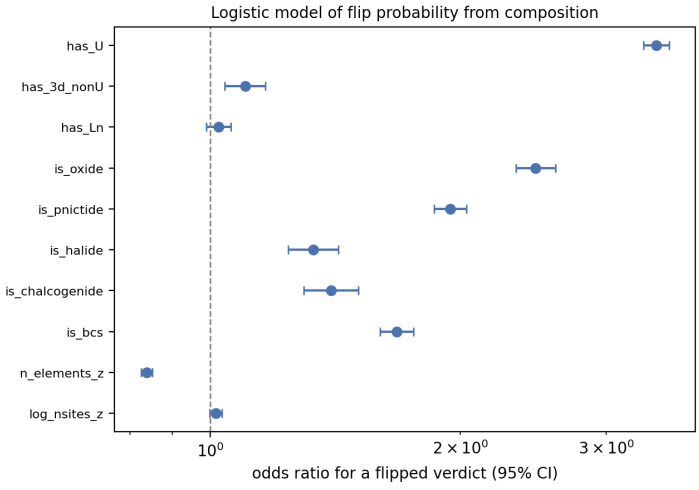


Figure 4: Odds ratios, log scale. Intervals plotted are naive; cluster-robust intervals (Table 7) are roughly 2.8 times wider and are the ones to read. The open-shell d term dominates under either; lanthanide content is indistinguishable from unity.

3.1, and at higher baselines it compresses further. Lanthanide content has an odds ratio of 1.02 with an interval straddling unity. And element count enters negatively — each standard deviation reduces the odds by about 16% — consistent with the marginal rates (binaries 19.8%, ternaries 14.6%, quaternaries 12.7%).

The cautions. First, the pseudo-R^2 is 0.066: composition ranks materials by risk well enough to be useful for allocation, but explains little variance in individual outcomes. Readers should not use this table to decide whether a specific compound's label is wrong. Second, correcting for clustering inflates every standard error by a factor between 2.0 and 3.5, so the naive intervals of an earlier draft were substantially overconfident; the open-shell d association survives the correction with a wide margin, but the halide, other-3d and log-sites terms now have intervals that include unity. Third, the open-shell d indicator aggregates eight chemically distinct elements and may be standing in for several underlying effects. We therefore describe it as the strongest observed predictor of functional disagreement in this dataset, not as an identified physical mechanism.

4.10 The lanthanide effect reverses sign with the anion

Our pre-registered expectation, following the standard concern about frozen-4f pseudopotentials, was that lanthanide materials would be least reliable. The concern is well founded in the literature: An et al. excluded 142 rare-earth compounds on exactly these grounds (An et al., 2026), GGA treatment of the Gd $4f^7$ shell has been argued to render the Materials Project hull unreliable for these elements (Tao et al., 2026), and frozen-f potentials are documented as appropriate only where f electrons are unimportant (van Setten et al., 2018).

On average the expectation fails: lanthanide materials flip at 11.8% against a global 13.0%. But the matched-cell comparison shows this average is misleading. Among quaternary 3d-containing materials, lanthanides *reduce* the flip rate in halides (9.1% against 22.4%) and oxides (18.1% against 30.7%), and *raise* it in pnictides (30.1% against 17.8%). Among quaternaries without a 3d metal, lanthanides raise the pnictide rate from 6.0% to 13.9%. The sign depends on the anion.

A flip in lanthanide chemistry is not evidence that r^{2}SCAN is correct: neither functional treats frozen-4f physics properly, and the disagreement is a flag, not a verdict.

4.11 Matched against known materials, the aggregate comparison inverts

GNoME materials flip at 13.0%; Materials Project entries computed at both levels flip at 14.9%. Taken at face value, AI-discovered materials appear *more* functional-robust, consistent with the source paper's comparison of its 84% against the Materials Project's 90%.

Twenty cells match between the two sets. GNoME is worse in 14, better in 6 (Table 8, Figure 5). The largest deficits are quaternary 3d oxides (+17.3 points), ternary 3d chalcogenides (+12.5) and ternary 3d borides/carbides/silicides (+11.3). The largest advantages are plain ternary intermetallics (−7.7) and ternary lanthanide intermetallics (−6.6).

The aggregate comparison inverts because GNoME's composition mix is dominated by intermetallics — metals exceed 60% of the dataset (Liu and Meng, 2024) — the chemistry in which it does well. A composition-weighted average therefore flatters it.

This comparison carries the weakest inferential warrant in the paper, and we state its limits plainly. One limit has a known direction. An incomplete r^{2}SCAN hull omits competing phases and so understates flip rates, on either side. On the Materials Project side we can measure completeness directly (Section 4.14): the r^{2}SCAN hulls of known oxides are the least complete of any anion class (median 37% of competing GGA entries recomputed) and those of intermetallics the most complete (64%). The known oxide flip rate is therefore the most understated, and the +17-point gap in 3d

Table 8: Matched cells, GNoME against Materials Project previously catalogued materials. Selected rows; positive Δ means GNoME disagrees more.

Cell	GNoME	Known	Δ (pp)
4-element oxide, $3d$, no Ln	30.7%	13.3%	+17.3
3-element chalcogenide, $3d$	30.3%	17.8%	+12.5
3-element boride/carbide/silicide, $3d$	29.6%	18.3%	+11.3
3-element halide, no Ln/$3d$	17.3%	9.2%	+8.2
4-element oxide, no Ln/$3d$	18.9%	10.8%	+8.2
3-element oxide, no Ln/$3d$	21.7%	16.7%	+5.0
3-element intermetallic, Ln, $3d$	19.0%	20.6%	−1.5
3-element halide, Ln	10.6%	14.8%	−4.2
3-element intermetallic, Ln, no $3d$	6.4%	13.0%	−6.6
3-element intermetallic, no Ln/$3d$	7.9%	15.7%	−7.7

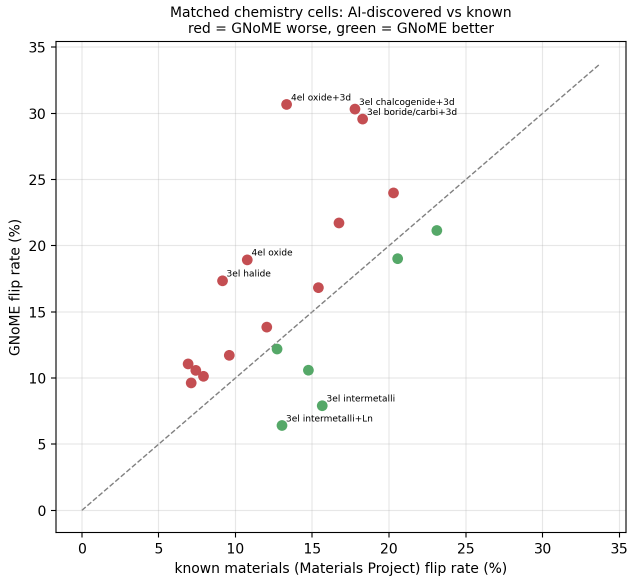


Figure 5: Flip rate in matched composition cells. Points above the diagonal are cells where GNoME materials disagree more than previously catalogued materials of the same chemistry.

oxides is a lower bound with respect to this bias. In intermetallics the Materials Project hull is comparatively complete while GNoME's is, by the "All"-reference evidence, the least complete of any class, so part of GNoME's −7.7-point advantage there may be incompleteness on the GNoME side rather than agreement. Matching the two sets on distance below the PBE hull, which Section 4.4 shows matters within GNoME, is not possible: the Materials Project reports `energy_above_hull` floored at zero for stable entries and exposes no sub-hull margin. We note only that the GNoME-side gradient holds among ternaries within 20 meV/atom of the hull, where the known-materials cells are concentrated: 11.6% for plain intermetallics ($n = 3{,}101$) against 38.5% for oxides ($n = 304$), 37.2% for $3d$ pnictides ($n = 400$) and 45.9% for $3d$ borides, carbides and silicides ($n = 292$). The two sets are matched on composition class and element count only. They differ in structural complexity, distance from the hull, provenance, phase-diagram completeness and the process by which each was selected for r^{2}SCAN recomputation. The defensible claim is therefore: *within the analysed datasets, functional-level disagreement differs between GNoME and previously catalogued materials in a chemistry-dependent manner, and the direction of the aggregate comparison differs from that seen in several matched composition cells.* It is not a claim that AI discovery as such makes a material's label more or less reliable, and we offer no confident explanation for the direction of the effect.

4.12 Chemistry-aware verification budgets

If flip rates are predictable from composition, a maintainer with a limited r^{2}SCAN budget should not sample uniformly. The policy is specified as follows, so that it can be reimplemented exactly. Half of the 8,429 prototype clusters are drawn uniformly at random without replacement to form the estimation set, and all materials in the remaining clusters form the evaluation set; **the split is by cluster, not by material**, so near-duplicate structures never straddle the two, and splitting by material would leak information and inflate the result. A flip rate is estimated for each (element count, anion class, lanthanide, $3d$) cell from the estimation set only. Cells absent from the estimation set are assigned that set's global rate. Evaluation-set materials are ranked by their cell's estimated rate in descending order; materials within a cell are tied and are ordered arbitrarily by a stable sort, which is immaterial because the catch rate depends only on how many members of each cell fall inside the budget. A budget b selects the first $\lfloor bN \rfloor$ materials. The procedure is repeated over 20 independent cluster-wise splits and we report the mean and standard deviation across them. We apply no shrinkage to small-cell estimates; doing so would plausibly improve the policy and we have not tested it.

Table 9: Share of flipped labels surfaced at a given recomputation budget. Mean over 20 cluster-wise splits, with standard deviation across splits.

Budget	Chemistry-ranked	s.d.	Uniform random	Difference
5%	11.4%	0.59	5.0%	+6.4 pp
10%	21.2%	0.65	10.0%	+11.2 pp
20%	37.1%	0.87	20.1%	+17.0 pp
30%	50.7%	0.75	30.1%	+20.6 pp
50%	70.9%	0.81	50.0%	+20.9 pp

At a 20% budget, chemistry-aware allocation surfaces 1.85 times as many disagreements, and the across-split standard deviation is under one percentage point, so the advantage is not an artefact of a lucky split. The operational rule is to prioritise $3d$-containing oxides, pnictides and chalcogenides

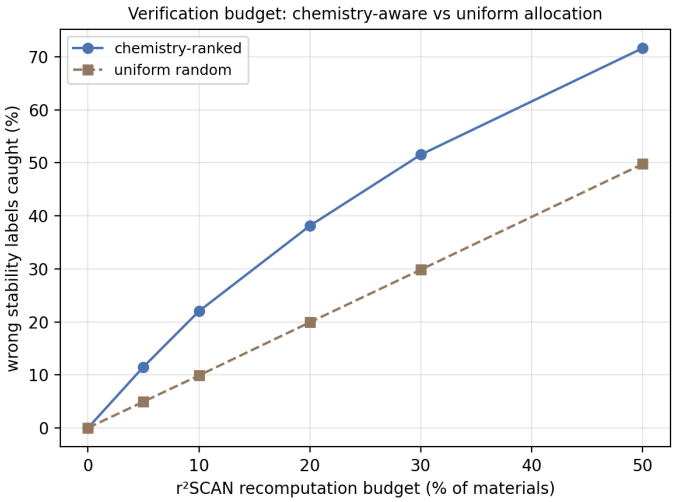


Figure 6: Flipped labels surfaced against recomputation budget, chemistry-ranked versus uniform random. Curve shown is a single split; Table 9 gives means over 20.

and deprioritise plain intermetallics and lanthanide halides; given the per-element results, an even simpler rule captures much of the benefit, namely recompute anything containing Fe, Co, Cr or Mn first.

This demonstrates that chemistry-aware ranking beats uniform sampling under the tested setup. It does not establish that this policy is optimal: incorporating hull distance, structural similarity or uncertainty in the estimated cell rates would plausibly improve it.

4.13 Robustness

Threshold. Flip rates were recomputed at thresholds from 0 to 50 meV/atom (Appendix C). The ordering of anion classes holds at every threshold, so the result is not an artefact of the zero cutoff. Reporting against threshold rather than at a single cut follows Reents et al. (2026).

Reference set. All conclusions hold against both the MP and MP+OQMD hulls; the latter runs 2–4 points higher uniformly, with cell ordering preserved. The "All" reference set behaves differently and is treated in Section 4.14.

Pseudopotentials. GNoME used newer PAW potentials for Li, Na, Mg, Ge and Ga. Materials containing those elements flip *less* (11.0% against 13.5%), so this difference does not generate the effect.

4.14 Hull completeness, probed on both sides

The release includes a third decomposition-energy column labelled "All" that is not defined in the accompanying documentation. Against it the global flip rate is 25.8%, roughly double the MP-referenced figure. A reviewer is entitled to ask why we did not take the larger number as primary.

Our reading, which we state as an inference and not a fact, is that "All" includes GNoME's own structures as competing phases, so that the reference hull is more complete than the MP hull alone. Two observations support this. A more complete hull can only raise decomposition energies, and

"All" raises them everywhere. And the chemistry that gains the most is intermetallics (Table 10), which is where GNoME added the largest number of new phases and where the MP hull is therefore least complete.

Table 10: Anion-class flip rate against the MP reference hull and the "All" reference set.

Anion class	n	MP reference	"All" reference
oxide	13,668	19.3%	29.5%
pnictide	26,330	16.7%	29.6%
boride/carbide/silicide	25,732	15.0%	29.6%
chalcogenide	7,849	12.1%	21.6%
hydride	1,827	12.0%	22.3%
halide	9,840	11.5%	24.2%
intermetallic	54,295	9.2%	22.3%

If this reading is right, the "All" column is a partial measurement of the hull-completeness bias on the GNoME side. The chemistry gradient survives it: oxides, pnictides and borides remain highest and intermetallics remain lowest. But the spread compresses, from a 2.1-fold ratio between oxide and intermetallic under the MP reference to 1.4-fold under "All." We therefore report the MP-referenced gradient as an upper bound on the chemistry effect and the "All"-referenced gradient as a lower bound, conditional on the inference above. We kept the MP reference as primary because it is the quantity the source paper reports and the one against which we validated the pipeline, not because it gives the larger spread.

The Materials Project side can be inspected directly. GNoME's hulls are not released, but the Materials Project's are. For each of the 22,756 known materials in Section 4.11 we define r^2SCAN hull completeness as the fraction of GGA($+U$) entries in the material's chemical system and all of its subsystems, excluding the material itself, that also carry an r^2SCAN thermo document. This is a coarse measure: it counts every entry, not only the ground-state structures that define the hull, so it understates the completeness that matters. It is nevertheless the first direct measurement of the quantity, and it varies strongly with chemistry (Table 11). Oxides have the least complete r^2SCAN hulls of any class and intermetallics the most complete, the reverse of what the GNoME "All" comparison suggests for GNoME's own hulls. Because incompleteness understates flips, the known oxide flip rate in Table 11 is the most understated and the intermetallic rate the least; with complete hulls the oxide–intermetallic ordering on the Materials Project side would widen, not close.

Table 11: Materials Project side: median r^2SCAN hull completeness and flip rate by anion class, for the 22,756 GGA($+U$)-stable materials with r^2SCAN data.

Anion class	n	Median completeness	Flip rate
oxide	4,065	37.0%	17.0%
intermetallic	7,953	64.1%	16.4%
pnictide	2,801	53.3%	15.2%
halide	2,634	58.2%	13.0%
boride/carbide/silicide	2,365	42.0%	12.1%
chalcogenide	2,704	58.5%	11.7%
hydride	233	67.3%	9.4%

The sharper test is whether the open-shell d effect survives at fixed completeness. Splitting the known materials into completeness terciles (cut at 46.2% and 61.1%), the flip rate for materials containing a $3d$ element is 21.5%, 19.8% and 18.7% in the low, middle and high terciles against 13.4%, 9.9% and 13.4% for materials without one: an excess of +8.1, +9.9 and +5.3 points that is not monotone in completeness and remains above +5 points in the most complete tercile. Within the lowest tercile, the only one in which both cells are populated, $3d$ oxides flip at 21.0% ($n = 1{,}205$) against 14.6% for plain intermetallics ($n = 315$). The $3d$ effect is not a completeness artefact on the side where completeness can be measured.

5 Selection effects

Beyond the propensity analysis of Section 4.7, two coverage features constrain what can be claimed.

Element count. Coverage is 12.9% for binaries, 31.8% for ternaries and 31.4% for quaternaries, then collapses to 0.13% for quinaries (229 of 177,392) and zero for senaries (0 of 4,792). The r^{2}SCAN validation effectively stops at four elements.

This is consequential. GNoME presents discovery in combinatorial spaces of four or more elements, and emergent generalization to five and six, as a central result (Merchant et al., 2023). Quinaries make up 32% of the released dataset. That regime has no meta-GGA validation. Our logistic model finds flip risk *decreasing* with element count, which would predict quinaries to be relatively safe — but extrapolating a fitted trend past the edge of the data is exactly the move this paper argues against, and we decline to make it.

Distance to the hull. Coverage declines monotonically toward the hull, from 26.1% in the $[-0.1, -0.05]$ eV/atom band to 16.5% in $[-0.01, 0]$ and 0.3% above zero. Supplementary Note 2 of the source work states that disagreement concentrates near the hull. The recomputed subset therefore under-samples precisely the region where flips are most likely, and 13.0% should be read as a lower bound on the population rate. The propensity model in Section 4.7 includes PBE decomposition energy as a covariate and so partially corrects for this.

Censoring bounds. Treating all unrecomputed materials as flipped, then as unflipped, gives bounds of 3.3% to 78.1% across the full 554,054-entry release. These are uninformative, and we report the IPW estimate of 12.7% conditional on the $\leq$4-element population instead.

Unmatched identifiers. 22,498 r^{2}SCAN material IDs do not appear in the current summary file. Their composition profile matches the remainder and their flip rate is similar (11.4% against 13.4%), consistent with a release-version mismatch. Including or excluding them changes no conclusion.

6 Discussion

For users of generated catalogues. A single reliability figure attached to a database is not usable. A group screening GNoME for transition-metal oxide cathodes faces roughly a 30% rate of verdicts that r^{2}SCAN would overturn; a group screening for intermetallics faces roughly 4%. We suggest datasets of this kind ship a per-entry disagreement flag rather than a global statistic.

For benchmark design. Benchmarks evaluate MLIPs against a PBE hull treated as ground truth (Riebesell et al., 2025). Our results imply the noise in that ground truth is chemistry-structured. A model that appears to underperform on transition-metal oxides may be partly disagreeing with labels that are themselves functional-sensitive in that chemistry. We do not claim this explains any particular benchmark result.

For MLIP training data. Warford et al. argue that selectively applied U splits a training set into incompatible surfaces (Warford et al., 2026). Section 4.3 shows the resulting label disagreement is concentrated in exactly the elements they identify. Datasets built at the r^{2}SCAN level from the outset, such as MatPES (Kaplan et al., 2025) and MP-ALOE (Kuner et al., 2025), avoid the issue.

What would settle the physics. Our evidence is statistical. We show that open-shell d content predicts disagreement after controlling for the correction scheme, band gap, structure, size and recomputation selection. We do not show *why* the two functionals disagree for these materials. Establishing that requires a targeted study of a small number of systems at higher levels of theory.

7 Limitations

Hull completeness is the largest partially resolved threat. An r^{2}SCAN convex hull omits any competing phase for which no r^{2}SCAN calculation exists, biasing toward apparent stability and therefore *understating* flip rates. The Materials Project's mixed scheme guards against this by using r^{2}SCAN formation energies only where every ground-state structure has been recomputed (Materials Project, 2026), but the pure r^{2}SCAN thermo type we query does not, and the GNoME decomposition energies we analyse were computed by the dataset authors against hulls we cannot inspect.

We state the consequence precisely. If reference-phase coverage varies systematically with chemistry, part of the gradient we attribute to functional disagreement could instead reflect differential hull completeness. Section 4.14 provides a partial measurement: under the more inclusive "All" reference set, which we infer to be more complete, the oxide-to-intermetallic ratio falls from 2.1 to 1.4 while the ordering is preserved. The gradient is therefore real but its magnitude depends on hull completeness, and we report the MP-referenced and "All"-referenced figures as upper and lower bounds. On the Materials Project side, where completeness can be measured, it varies by anion class in the direction that understates oxide flip rates most, and the open-shell d excess is unchanged across completeness terciles. A direct test on the GNoME side would require the dataset authors' hull construction, which is not released.

Neither functional is ground truth. r^{2}SCAN is more accurate than PBE for formation energies on average (Kingsbury et al., 2022a), but not uniformly: for transition metals and intermetallics the two are comparable and PBE is sometimes better (Kothakonda et al., 2023; Bartel et al., 2019), and for copper SCAN and r^{2}SCAN overstabilise the Cu^{2+} state relative to Cu^{+} where PBE+U reproduces experiment (Bae et al., 2025). A low flip rate for a chemistry, as we find for copper, therefore says only that the two functionals agree there, not that either is right. The noise floor is substantial: experiment-versus- experiment MAE is 0.082 eV/atom (Kirklin et al., 2015), and MP2020 correction uncertainties alone can move near-hull polymorphs across the boundary (Wang et al., 2021).

Clustering is corrected but the proxy is imperfect. Our prototype proxy is (space group, anonymised stoichiometry). This will merge genuinely distinct structures sharing a space group and stoichiometry pattern, and will split related structures that relaxed into different space groups. A true prototype assignment, for instance via a structure-matching algorithm, would be better. We mitigate this by showing the headline result is unchanged under a coarser definition (Section 3.3), and note that because merging inflates cluster size, our design effect is more likely an over- than an under-correction.

Composition explains little individual variance. Pseudo-$R^2 = 0.066$. The model supports population-level triage, not per-material prediction.

The known-materials baseline is itself selected. Four-element known cells are small ($n =$ 108–297), and "known" means previously catalogued, including theoretical entries.

Unexplained features. Nickel's position at the global average despite receiving U, and the triclinic excess, are both unexplained.

Scope. Every number applies to GNoME materials with at most four elements that were selected for r^{2}SCAN recomputation.

8 Conclusion

A single aggregate survival rate is not an adequate description of how reliable AI-generated stability labels are with respect to a higher-fidelity functional. Resolved by chemistry, the PBE→r^{2}SCAN flip rate varies more than sevenfold across composition cells and, among non-f elements, roughly sevenfold across elements. The variation survives correction for non-independence among substitution-generated structures, stratification by distance to the PBE hull, controls for band gap and structure, and reweighting for recomputation selection; it compresses but persists under a more complete reference hull. It is produced by a directional shift, not by noise: r^{2}SCAN moves open-shell $3d$ oxides and pnictides toward instability and plain intermetallics toward stability. The empirical Hubbard-U correction does not generate the effect and partially suppresses it where applied, so the largest disagreements in the dataset are open-shell d elements in chemistry that receives no correction. Open-shell d content is the strongest observed predictor of disagreement across every specification we tested; we do not claim to have identified the physical mechanism, and establishing one would require targeted higher-level calculations on a small number of systems. Lanthanide content, the usual suspect, is not significant on average and reverses sign with the anion. The comparison against previously catalogued materials reverses direction depending on composition. Verification budgets should follow the chemistry, and the compositional regime GNoME advertises most prominently remains unvalidated.

Data and code availability

All input data are public: the two GNoME CSV files are served from the DeepMind bucket `gs://gdm_materials_discovery/gnome_data/` (139,541 and 554,054 rows at the time of access, 9 October 2026), and the Materials Project thermo records were retrieved through the public API on 9–10 October 2026 with the queries recorded in the scripts. The repository at https://github.

com/FOUNDERACCRETION/gnome-audit contains the pre-registered decision rules with their commit history, one script per analysis phase, every machine-readable table and figure the paper draws on, pinned package versions, and a `run_all.py` driver that regenerates the results directory from the two CSV files and checks twenty quantities against the values printed here: every number in the abstract, and the within-anion excess table of Section 4.3. A continuous-integration workflow performs that regeneration on a clean runner on every push that touches the analysis code, after deleting the stored results so that the check cannot pass on them, and fails if any phase crashes even when the surviving numbers match. The only components not reproducible without credentials are the two Materials Project pulls, which require a free API key.

Disclosures

AI assistance: Claude (Anthropic) was used throughout. It proposed the research question from a shortlist, carried out the literature search, drafted the pre-registered decision rules, wrote and executed every analysis script, drafted the manuscript, and produced the adversarial referee reports to which later drafts respond. Bibliographic details were checked by Claude against publisher and arXiv records; the author has not independently re-verified every reference. The author chose the question, committed the decision rules before the stratified analysis was run, obtained the data and the Materials Project credentials, ran the Materials Project queries, reviewed the outputs, and is responsible for the claims made. Competing interests: the author is the founder of Accretion Labs, a materials-informatics company.

A Cell-level results with cluster-robust intervals

All disjoint cells with $n \geq 500$. "Naive" intervals treat materials as independent; "cluster" intervals resample prototype clusters. The final column is the ratio of interval widths.

n_{el}	Anion	Ln	$3d$	n	Rate	Cluster 95% CI	Widen
4	oxide	no	yes	2478	30.7%	[27.6, 33.8]	1.72
3	chalcogenide	no	yes	564	30.3%	[25.3, 36.7]	1.49
4	pnictide	yes	yes	4573	30.1%	[26.1, 33.7]	2.76
3	boride/carbide/silicide	no	yes	575	29.6%	[24.2, 35.4]	1.43
4	boride/carbide/silicide	no	yes	4501	25.2%	[21.2, 29.3]	3.32
3	pnictide	no	yes	813	24.0%	[19.7, 28.6]	1.49
4	boride/carbide/silicide	yes	yes	4561	22.8%	[18.9, 27.6]	3.77
4	halide	no	yes	794	22.4%	[17.4, 27.4]	1.69
3	oxide	no	no	921	21.7%	[18.6, 24.7]	1.14
3	intermetallic	no	yes	2667	21.1%	[18.4, 24.0]	1.79
3	intermetallic	yes	yes	2056	19.0%	[15.9, 22.4]	1.96
4	oxide	no	no	3318	18.9%	[16.8, 21.4]	1.69
4	oxide	yes	yes	1399	18.1%	[14.2, 22.0]	1.93
4	pnictide	no	yes	5736	17.8%	[14.6, 20.8]	3.33
3	halide	no	no	1326	17.3%	[14.4, 20.9]	1.68
3	pnictide	yes	no	1349	16.8%	[12.0, 22.3]	2.43
4	chalcogenide	no	yes	1487	16.3%	[13.4, 20.0]	1.74
4	pnictide	yes	no	7412	13.9%	[11.6, 16.5]	3.19

n_{el}	Anion	Ln	$3d$	n	Rate	Cluster 95% CI	Widen
4	oxide	yes	no	4232	13.8%	[11.1, 16.3]	2.73
4	halide	no	no	1404	13.1%	[10.0, 16.3]	1.84
4	intermetallic	yes	yes	10659	12.9%	[11.7, 14.5]	2.19
3	oxide	yes	no	730	12.2%	[8.0, 16.6]	1.78
4	intermetallic	no	yes	7459	12.1%	[10.7, 13.5]	1.96
3	boride/carbide/silicide	no	no	845	11.7%	[8.9, 14.7]	1.32
3	pnictide	no	no	1473	11.1%	[9.0, 13.5]	1.38
3	halide	yes	no	982	10.6%	[6.6, 16.2]	2.60
3	chalcogenide	no	no	1106	10.6%	[8.3, 12.9]	1.32
3	chalcogenide	yes	no	1076	10.1%	[7.0, 14.0]	1.96
3	boride/carbide/silicide	yes	no	1246	9.6%	[6.3, 13.7]	2.37
4	halide	yes	yes	1119	9.1%	[5.6, 14.2]	2.74
4	boride/carbide/silicide	yes	no	10287	9.1%	[6.9, 11.4]	4.31
4	chalcogenide	yes	no	1497	8.4%	[6.2, 10.7]	1.60
4	boride/carbide/silicide	no	no	3422	8.2%	[6.8, 9.7]	1.55
3	intermetallic	no	no	2618	7.9%	[6.6, 9.2]	1.20
4	chalcogenide	no	no	1467	7.0%	[5.3, 8.7]	1.20
3	intermetallic	yes	no	4661	6.4%	[5.2, 8.1]	2.13
4	pnictide	no	no	4675	6.0%	[4.8, 7.2]	1.79
4	halide	yes	no	3599	5.9%	[2.8, 10.8]	4.65
4	intermetallic	yes	no	18932	5.5%	[4.6, 6.3]	2.63
4	intermetallic	no	no	5025	4.0%	[3.2, 4.8]	1.44

Median widening factor 1.82. The highest and lowest quaternary cells remain disjoint: 30.7% [27.6, 33.8] against 4.0% [3.2, 4.8].

B Per-element results

All 83 elements with $n \geq 300$, ordered by flip rate. Point estimates; cluster-robust intervals for the extremes are given in Section 4.2. All 83 contrasts against the global rate survive Benjamini–Hochberg correction at 0.05.

El.	n	Flip
Fe	5762	42.8
Co	10972	33.2
Cr	2553	30.7
Mn	4994	25.2
F	2801	24.9
As	6298	21.6
Ce	4236	20.5
Mo	3910	20.2
Sb	6577	19.9
O	13668	19.3
V	3659	18.9
Si	18382	18.4
W	3223	18.2
Eu	2132	18.0
U	2595	17.9
Te	2207	17.4
Bi	6606	16.5
Ag	4224	16.3
Gd	976	16.0
Zr	7315	14.7

El.	n	Flip
Li	6305	12.2
Ba	3580	12.1
Tb	12248	12.1
Nb	5297	11.9
Tm	10154	11.8
Er	10434	11.6
Na	2932	11.6
Ta	5021	11.3
Nd	9238	11.1
La	6546	11.0
Rb	3141	10.8
Pu	1020	10.8
Br	3440	10.6
Pb	4377	10.6
B	9496	10.6
Hg	4445	10.2
Cs	2973	10.1
Th	7632	9.9
Al	11933	9.7
Np	1387	9.7
Zn	7510	9.1
Cd	5356	9.0
In	8259	9.0
Os	8652	8.9
N	3845	8.7
Pd	10381	8.1
Re	3958	8.1
Ru	12381	8.1
Sm	4852	8.0
I	3091	7.9
Pt	13450	7.8
Ge	4764	7.6
Rh	14340	7.3
Yb	6265	7.1
Be	2799	6.5
Ir	15680	6.2
Tc	7350	5.7
Mg	4073	5.7
Pm	389	4.4
Pa	4251	3.6
Ac	2181	1.4

C Flip rate against threshold

Pnictides, omitted for width, run from 16.7% at zero threshold to 2.2% at 50 meV/atom.

Thresh. (eV/at)	Overall	B/C/Si	Chalc.	Halide	Hydride	Interm.	Oxide
0.000	13.0%	15.0%	12.1%	11.5%	12.0%	9.2%	19.3%
0.005	9.3%	11.3%	8.0%	8.2%	8.6%	5.5%	15.5%
0.010	7.2%	8.6%	6.0%	6.4%	6.0%	3.9%	12.5%
0.020	4.5%	5.2%	3.8%	3.6%	4.3%	2.3%	8.6%
0.030	3.0%	3.5%	2.6%	2.2%	2.5%	1.5%	6.1%
0.040	2.1%	2.6%	2.0%	1.5%	1.7%	1.0%	4.2%
0.050	1.6%	2.1%	1.6%	1.2%	1.1%	0.8%	3.2%

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